

Final Team Proposal for Move Tickers

Company Risk Analysis and Strategy Simulator

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Block 1D – 2024–2025

28 May 2025

Introduction

Beating the stock market is notoriously difficult, even for experienced investors. If even the experts struggle, how is the average consumer supposed to keep up? With the rise of machine learning and increased access to financial data, we now have the tools to analyse both past and present trends, uncover patterns, and make educated predictions. Move Tickers aims to bridge that gap by providing AI-powered tools that support strategic decision-making and help users anticipate market changes with greater confidence. By doing so, it empowers everyday investors to face financial uncertainty with the kind of insight that was once only available to professionals.

This project aims to develop a machine learning model that is capable of:

- **Classifying** companies from the NASDAQ-100 index into -risk tiers (Low, Moderate, High) based on their volatility and sensitivity to macroeconomic indicators.
- **Predicting** short-term market direction (i.e., whether the market is likely to move up or down in the next day or week) using key macroeconomic variables such as CPI and volatility indices

These components will be built into an interactive web interface that will allow users to explore company-specific risk profiles, test a particular trading strategy, and gain insight into potential upcoming market shifts based on real-world economic trends. By combining macro-sensitivity clustering with predictive modelling, the project aims to deliver actionable insights tailored to the

specific needs of investors. Therefore, this project contributes to Move Tickers' goal of making advanced financial insights more accessible to the public.

Problem Statement

Move Tickers currently lacks an intelligent and data-driven system that is capable of assessing macroeconomic risk at the scope of individual company level and forecasting significant shifts in the stock market. Without such a system in place, everyday users are left without the tools they need to make informed decisions, especially in a market filled with countless options and high risk of costly mistakes.

To fill this gap, the project focuses on building an application capable of:

- **Segmenting** companies based on their historical volatility and macroeconomic sensitivity
- **Forecasting** short-term market direction using key macro indicators

To meet Move Tickers' need for a tool that simplifies the complex market data and making it easily accessible for users, the models will be integrated into an interactive application showcasing six representative NASDAQ-100 companies, each with a distinct risk profile. The interface will guide users in selecting companies that align with their individual risk–return preferences, helping them build strategies that support their investment goals.

The project is guided by two key questions:

- *Can companies be meaningfully **segmented** by their volatility and **macroeconomic** behaviour?*
- *How accurately can **macroeconomic** indicators predict short-term market direction?*

By addressing these questions, the project aims to provide data-driven solutions for the users of Move Tickers.

Data Description

The data for this project comes entirely from the dataset provided by Move Tickers. This dataset includes structured tables that cover data from companies in the NASDAQ-100 but also about oil, gold and macroeconomic data from the past years.

While several datasets are used, a significant part of the analysis draws from the daily dataset, which provides valuable data on price and volume data over a multi-year period. This like term perspective is essential for identifying consistent trends like measuring historical volatility and or assessing how companies coped with macroeconomic shifts over time.

For example, classifying a company like Apple as low-risk is only meaningful when backed by stable performance data spanning several years, particularly through periods of economic uncertainty. Likewise, high-risk companies such as Palantir require observing recurring volatile behaviour in response to technology advancements or macroeconomic events. To observe this effectively, we, in addition to the daily dataset, also looked at the macroeconomic dataset, which includes indicators such as CPI (Consumer Price Index, a common metric used to measure inflation) and the interest rate. Analysing the performance of the six selected companies alongside these macro variables allows us to create evaluations of how each company reacts under different economic conditions. By looking at a time frame of ten years, the ability to make evidence-based distinctions between low, moderate, and high-risk profiles is greatly strengthened. As a result, the classification becomes more reliable for users intending to use these models to make investment decisions.

Additionally, a dedicated table containing six selected NASDAQ-100 companies was created, each representing a distinct macro-risk profile: low (Microsoft, Apple), moderate (Qualcomm, KLA), and high (CrowdStrike, Palantir). These classifications are supported by historical data reflecting each company's volatility patterns and sensitivity to macroeconomic shifts. The resulting labels are used both to train the classification model and to provide users with clear, interpretable insights in the dashboard.

Data cleaning and feature engineering were handled using SQL to create tables, filter and format data as well as aligning the data about stocks with the macroeconomic indicators. Furthermore, modelling and analysis steps were performed in Python to prepare the dataset for the classification and prediction component of the project.

Methodology

Data Collection & Pre-processing

How will data be gathered?

The data was provided by Move Tickers in the form of CSV files, each representing daily stock price data for NASDAQ-100 companies. These files were accessed from the university server. Either directly on campus or remotely via VPN, and loaded into the Python environment for further processing and feature engineering.

How will data be cleaned?

Each dataset was parsed to ensure that dates were correctly formatted in chronological order. The data was checked for duplicate entries and rows and any missing values were also removed. Columns were renamed and standardised for consistency.

How will missing values or bias be dealt with?

Missing values were dropped in the initial cleaning step. The macroeconomic indicator data were only available in monthly intervals so forward imputation will be applied to align them with daily stock data. This is done to prevent gaps in the data while maintaining realistic inputs for daily level analysis.

Model Selection & Training

The project aims to produce a combination of clustering and classification models. For the macro risk classification, a **K-Means clustering** was used in order to group companies based on their historical volatility and behaviour. These clusters were then labelled and used in a **K-Nearest Neighbours (KNN)** to assign the 3 unique risk levels (High, moderate, Low) to the companies.

For the strategy testing and forecasting of the market trends, random forests classification models and time series based approaches were used since these are suitable for financial data. The model specs may differ depending on the behaviour of the data but may include: tree based classifiers or simpler linear models that can output short term predictions. Model experimentation is expected as the outcomes of the models are evaluated.

How will the data be split?

The plan is to use a time-based split on the data, training the models on the first seven years of available data and testing them on the most recent three years. By following this procedure, the chronological nature of the financial time series is respected, and any potential data leakage from the future into training is avoided.

Evaluation Metrics

Multiple evaluation metrics will be used to assess the performance of different models. These include:

- **Accuracy, Precision, Recall, and F1-score** for classification tasks
- **Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE)** for regression and prediction accuracy
- **Silhouette score** for clustering quality

The chosen metrics are intended to provide a comprehensive view of model performance. They are commonly used to evaluate classification, regression, and clustering models, making them a reasonable fit for this project. While their exact relevance will depend on the final models selected, starting with a broad set of metrics offers a solid foundation for comparison and evaluation.

Deployment

The final application aims to combine the models into one, in the form of an interactive web dashboard that will be built using Streamlit. It will facilitate the use of the models without requiring any technical knowledge of machine learning and will allow users to explore company-specific macroeconomic risk profiles, test strategies, and view the model’s trend predictions.

The dashboard will be hosted on the university server (BUas) as a standalone prototype. While full integration into Move Tickers’ system isn’t planned at this stage, the current setup is flexible enough to allow for future expansion or changes, depending on feedback and how the project develops.

Project Timeline

Break the work into phases:

Phase	Task	Team Member(s)	Deadline	Deliverable
1	Data Collection	Pol	Week 3	Cleaned dataset
2	Data Pre-processing	All of Group	Week 3-4	Feature set
3	Model Training	All of Group	Week 5	Trained model
4	Model Evaluation	All of Group	Week 5-6	Evaluation report
5	Deployment/Presentation	All of Group	Week 6-7	Final app/report

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